

Parsa Kafashi

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Research interests

- Quantum Complexity in Quantum Gravity
- Quantum Chaos and Information Scrambling
- Applications of Quantum Information Concepts to High-Energy and Many-Body Physics
- Holography and emergent space-time

Education

Sharif University of Technology, Tehran, Iran

2024 – Present

M.Sc. Physics

- Ranked 9th out of more than 20,000 in Iran's National Master's Physics Entrance Exam (2024)

Sharif University of Technology, Tehran, Iran

2019 – 2024

B.Sc. Physics

- GPA: 17.27/20.00
- Ranked among the top 1% in Iran's National University Entrance Exam (2019)

Helli High School (NODET), Tehran, Iran

2016-2019

Diploma in Mathematics and Physics

- NODET: National Organization for Development of Exceptional Talents

Research Experience

Krylov Complexity for time-dependent generators

2026

- Studied the growth of Krylov complexity in quantum systems with time-dependent Hamiltonians. Developed a framework applicable to both periodically driven and general non-periodic dynamics, introducing reliable approximation methods to track complexity evolution even when exact solutions are unavailable.
arXiv:2607.10454 

Local Renormalization Group as Quantum Complexity

2025

Graduate thesis under the supervision of Dr. A. F. Astaneh

- A new notion of quantum complexity based on the local renormalization group is introduced. The physical intuition behind this definition is illustrated, and the formalism is formulated to be general across all dimensions and field theories. Its compatibility with path integral complexity in two dimensions is demonstrated. As an explicit example, the quantum complexity and optimized spacetime geometry are derived for a free massless scalar field in both 2 and 4 dimensions, as well as for a self-interacting scalar field in 2 dimensions. The optimization is performed by minimizing a specifically defined functional within the formalism.

Quantum-Compatible Approach to Complex System Dynamics

2025

- Conceived and developed a quantum information-inspired framework for analyzing multivariate time series, encoding system states as density matrices to capture higher-order correlations. Leveraged fidelity and entropy measures to quantify interdependencies, resilience, and recovery timescales without dimensional reduction. Validated the approach on synthetic data from a 9-dimensional Lorenz-96 model and applied it to real-world climate data to analyze global temperature anomalies across nine regions. This work was conducted in collaboration with Dr. M. R. Rahimi Tabar, Dr. A. Rezakhani, and other contributors.
arXiv:2512.14169 

Fast Radio Bursts as Cosmic Lightning

2025

- Developed a model attributing FRBs to sudden discharges of accumulated electric charge in the accretion disks of compact objects, such as black holes. Investigated how Compton scattering in the disk plasma generates charge separation and forms a capacitor-like system stabilized by radiation pressure and electrostatic forces. Analyzed the discharge mechanism leading to radiative emission, and compared predictions with

observational data (FRB2018725A) to extract quantitative relationships. Estimated total charge buildup and corresponding electron densities necessary for this process in stellar-mass black holes. Supervised by Dr. S. Rahvar.
arXiv:2508.03620 

Exploratory Research

Krylov Complexity in Curved Space-time and Fast Scrambling

2024-2025

- Explored the notion of Krylov complexity in a curved spacetime background by formulating a modified Hamiltonian for the Ising model that encodes geometric effects. Investigated the growth of Lanczos coefficients as a probe of information scrambling, with the aim of demonstrating how curvature can induce fast scrambling. This project inspired the idea of interpreting the local renormalization group as a manifestation of quantum complexity.

Large language models as $O(n)$

2024-2025

- Collaborated directly with Dr. Haghighat and his student at Tsinghua University, building on their work reformulating Transformer architectures as $O(N)$ models. Followed and discussed a novel idea to estimate the internal dimension of the model, gaining hands-on experience with large language models and Transformers, as well as insights into scaling behavior, phase transitions, and connections to the Ising model.

Drag Force in AdS/CFT

2023

- Studied and extended the work of Gubser on drag forces in $N=4$ Super-Yang-Mills plasma using AdS/CFT and classical string approximations. Investigated how extrinsic geometries affect the drag force on an external quark, gaining deeper understanding of holographic methods, thermal gauge theory, and applications to jet quenching phenomena.

Teaching Experience

Teaching Assistant, Advanced Electrodynamics

2026

- Instructor: Dr. Mehdi Kargarian
- Sharif University of Technology

Teaching Assistant, Introduction to Field Theory

2023

- Instructor: Dr. Amin Faraji
- Sharif University of Technology

Teaching Assistant, Gen Physics I

2024

- Sharif University of Technology

Instructor, Physics Olympiad: Electrodynamics and Mechanics

2020-2022

- Taught advanced problem-solving techniques to high school students for preparation of Iran's Physics Olympiad.

Selected Courses

- **String Theory** (Dr. A. F. Astaneh), **Introduction to String Theory** (Dr. H. Arfaei), **Quantum Field Theory** (Dr. N. Sadooghi), **Advanced Quantum Mechanics** (Dr. N. Sadooghi), **General Relativity** (Dr. S. Rahvar), **Selected Topics on Stochastic Physics** (Dr. M. R. Rahimi Tabar)

Technical skills

Programming Python, C++, Qiskit, Experience working with Large Language Models, Mathematica, $\text{\LaTeX}$
Languages: English(TOEFL iBT 110), Persian(Native), Japanese(JLPT N3)